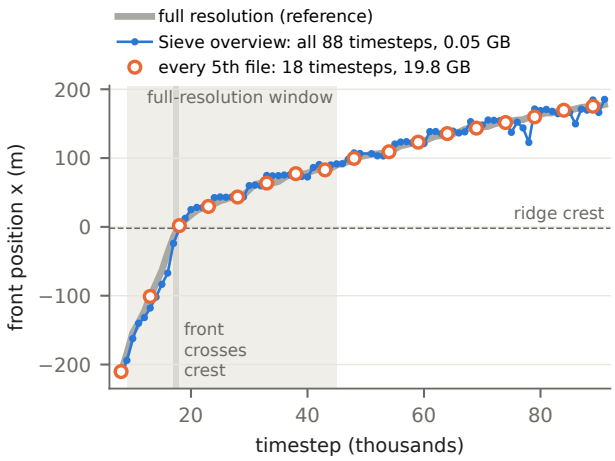



Figure 1: The fire front's position as each route sees it. Gray: full resolution at every timestep (a reference no route moves). Blue: Sieve's overview, every timestep at every eighth grid point (0.05 GB). Orange rings: every fifth file, a two-pass download's first pass (19.8 GB). Shaded: the window then fetched at full resolution; dark band: the 1,000 steps in which the full-resolution front crosses the crest.

| | GB over the network | | | Bytes | Server | Lines |
|---|---|---|---|---|---|---|
| Route | overview | box | total | vs Sieve | time (s) | of code |
| Sieve | 0.05 | 0.36 | 0.40 | 1.0× | 804 | 9 |
| Scripts on the server | 0.05 | 0.35 | 0.40 | 1.0× | 719 | 111 |
| Two-pass download† | 19.76 | 32.94 | 52.70 | 131× | — | 99 |
| Download everything | 96.62 | 0 | 96.62 | 240× | — | 91 |

†Its overview holds 18 timesteps (every fifth file); every other route's holds all 88.

Table 1: What each route moves over the network to deliver the same two results: the overview, and the window and box at full resolution. Sieve's and the scripts' outputs are identical, bit for bit. Server time is the time spent on the server reading and cutting; reads on the server are examined in Sec. 4.3.

## 4 Usage Scenarios

### 4.2 Wildfire: Choosing the Time Window from an Overview

#### 4.2.1 Data and Analysis Requirements

The dataset is a FIRETEC simulation [2] of a fire spreading over a ridge. It is stored on a remote server as 88 VTK files, one per saved timestep, each 1.1 GB (96.6 GB in total). Each file holds nine variables on a $600 \times 500 \times 61$ terrain-following grid that lists the position of every grid point. The wind blows across the ridge and the fire spreads with it: up the windward slope, over the crest at $x \approx -2$ m, and down the lee slope. Time is given in simulation steps; the files do not record the length of a step.

Scientific Task In coupled fire–atmosphere simulations, the air just ahead of a fire can reverse and flow back toward it [1]. Dupuy et al. find this *indraft* in the lowest grid cell, typically within 10 m of the fire; only when the fire spreads downslope does it reach farther, up to 30 m ahead in shrubland and 40 m in grassland under a weak wind. In this run, the fire slows once it crosses the crest. The scientist asks whether, as it slows, the wind near the ground ahead of the front reverses and blows against the spread.

Answering this takes two kinds of data. The first is *when* the fire crosses the crest and slows. The series must be sampled densely

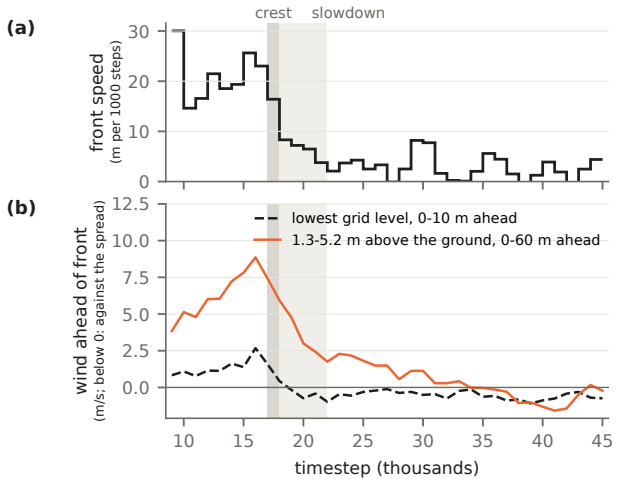



Figure 2: The answer, from the full-resolution window. Values are averaged over the burning strip ($|y| \leq 50$ m). (a) Front speed. (b) Wind along the direction of spread, ahead of the front: in the lowest grid level within 10 m of the front (dashed), and 1.3–5.2 m above the ground within 60 m of it (orange); below zero, the air blows back toward the fire. Light shading: the slowdown; dark band: the 1,000 steps in which the front crosses the crest.

| Sieve request | Estimated (GB) | Sent (GB) |
|---|---|---|
| Overview (all 88 timesteps) | 0.15 | 0.05 |
| Box (37 timesteps) | 0.46 | 0.36 |

Table 2: Sieve's estimate of each request, given before anything moves, against what it sent. The estimate counts the arrays uncompressed; Sieve compresses what it sends. The box's estimate is sized on the grid positions the overview brought back.

enough to time the drop, and how densely is not known in advance. The second is the wind in the lowest meters, near the front. An indraft lives in the lowest grid levels and the first meters ahead of the fire, so this needs the grid's full resolution, but only around the slowdown and only near the front.

Without Sieve, the scientist can copy the whole series; download every fifth file to find the slowdown, then the files of the chosen window; or write scripts that cut both on the server and copy the cuts back (Tab. 1). With Sieve, the scientist follows the visual information-seeking mantra, overview first, then zoom, filter, and details on demand [3], across the network and over time. From ParaView, through Sieve's plugin (Sec. 3.1.7), the scientist (1) requests an overview that keeps every timestep at reduced resolution (Listing 1); (2) chooses the time window and draws the box on that overview, with nothing moved; and (3) requests the window and the box at full resolution (Fig. 3). Each request is priced before anything moves, runs next to the data, and returns only its result, which ParaView opens in place. The measurements below are from re-running the two requests, as listed, from an empty cache.

```
src = source("ssh://host/wildfire/") # 88 files
view = fields(src, ["rhof_1", "theta", "u", "v", "w"])
view = subsample(view, 8)
save(view, "overview.vts")
```

Listing 1: Step 1: the overview. Every timestep, every eighth grid point along each axis, five fields.

#### 4.2.2 Step 1: An Overview of Every Timestep

The overview keeps every eighth grid point along each axis ($75 \times 63 \times 8$ points per timestep) and the five fields that follow the fire: the wind (`u`, `v`, `w`), temperature (`theta`), and the fuel left (`rhof_1`). Sieve returned all 88 timesteps in 47 MB, 0.05 % of the series. Downloading every fifth file instead moves 19.8 GB, about 420 times as much, for 18 timesteps.

The overview follows the front through the whole run (Fig. 1). Played in ParaView, it shows the fire slowing once it crosses the ridge. To time this, we ran a separate script on the overview Sieve returned: in each timestep it takes the front as the farthest point downwind where the lowest grid level is at least 400 K, within 50 m of the centreline, and divides the distance the front moves by the steps between timesteps. The same front can be tracked inside ParaView with its Threshold and Plot Data Over Time filters; Sieve supplies the data, not the analysis. On the overview alone, the front advances at 21 m per 1,000 steps on the windward slope, reaches the crest between steps 18,000 and 19,000, and moves at 2.8 m per 1,000 steps on the lee slope; the drop happens between steps 17,000 and 22,000. Every fifth file happens to include step 18,000, at the crest, so here it brackets the drop; with another offset, one of its 5,000-step gaps would hide it. Sampling the files risks missing exactly the moment the scientist is looking for. The overview carries no such risk: it keeps every timestep and thins only the grid. Without Sieve, getting it means logging in to the server, writing and running a script that cuts every file, copying the results back, and loading them into a visualization tool; with Sieve it is one four-line request (Listing 1), issued from ParaView.

*Is the overview enough to find the slowdown?* We compared it with the front computed at full resolution for every timestep, in a separate evaluation pass on the server. At every eighth point, the overview finds the fire at all 83 timesteps that have one; its front lies a median of 4.4 m from the full-resolution front (within one 16 m grid spacing at 92 % of timesteps), and it places the crest crossing one timestep (1,000 steps) late. Every fourth point gains 3 m of accuracy for eight times the bytes. Every sixteenth point misses the fire at 14 of the 83 timesteps and places the crossing two timesteps late. Every eighth point is the coarsest overview that still answers *when*.

The overview cannot answer the second question. Below 12 m above the ground it keeps two grid levels, the lowest and one at 11.8 m, of the nine the grid has there. In the lowest level, from the front to 60 m ahead of it, the overview's wind turns slightly negative from step 19,000, just after the crest crossing, while the level above stays near zero. With no levels between the lowest and 11.8 m, the overview cannot tell whether this reversal is confined to the lowest level or fills the first meters of air the fire spreads into.

#### 4.2.3 Steps 2 and 3: The Window and the Box at Full Resolution

From the overview, the scientist chose the window from step 9,000 to 45,000 (37 timesteps): it starts 8,000 steps before the drop and runs through the slow lee phase. On the overview in ParaView, they drew a box over the crest and the lee slope (Fig. 3); the plugin turned it into a `region` in the data's own coordinates. On this terrain-following grid the box holds $151 \times 121 \times 39$ points per timestep, with all nine grid levels below 12 m. Before fetching, Sieve estimated the request's size, using the grid positions the overview had brought back (Tab. 2); it then returned the 37 timesteps at full resolution in 0.36 GB. A two-pass download copies the window's files whole: 32.9 GB. The overview and the box are both saved with the position of every grid point, so the box sits inside the overview in ParaView without any alignment step (Fig. 3).

#### 4.2.4 Results

Figure 2 answers the question. During the slowdown, between steps 17,000 and 22,000, the front's speed falls from 23 to 4 m per 1,000 steps. Over the same steps, the air 1.3–5.2 m above the ground within 60 m ahead of the front weakens from +8.9 to +1.8 m/s, but keeps blowing with the spread until step 34,000. Only the lowest grid level reverses during the slowdown: within 10 m of the front it is negative from step 19,000, while the levels above it still blow forward at 4 to 7 m/s. This is the shallow indraft Dupuy et al. describe [1]. The overview, which keeps two of the grid's nine levels below 12 m, held the lowest and so hinted at this reversal; only the full-resolution box, with all nine, shows that the reversal is confined to the lowest level.

The slowdown thus coincides with the near-ground wind ahead of the front *weakening*, not reversing. The analysis shows when each happens, not what causes the slowdown. Before the fire arrives (steps 2,000 to 5,000), the near-ground air over the lee slope already flows backward at 2 to 6 m/s, so the terrain alone can reverse it. The height of the lowest grid level above the ground is not recorded in the files; the heights above are measured from it.

Every route in Tab. 1 delivers the same two results. Sieve and the scripts on the server move the same 0.40 GB, and their outputs are identical, bit for bit. The scripts take 111 lines on two machines: cutting next to the data, copying the cuts back, and rebuilding the VTK series locally, after logging in to the server. Sieve's two requests take 9 lines, which the plugin writes. On the server, Sieve's requests took 804 s and the scripts 719 s. The two-pass download moves 131 times Sieve's bytes and sees only 18 timesteps before choosing the window; copying the whole series moves 240 times as much.

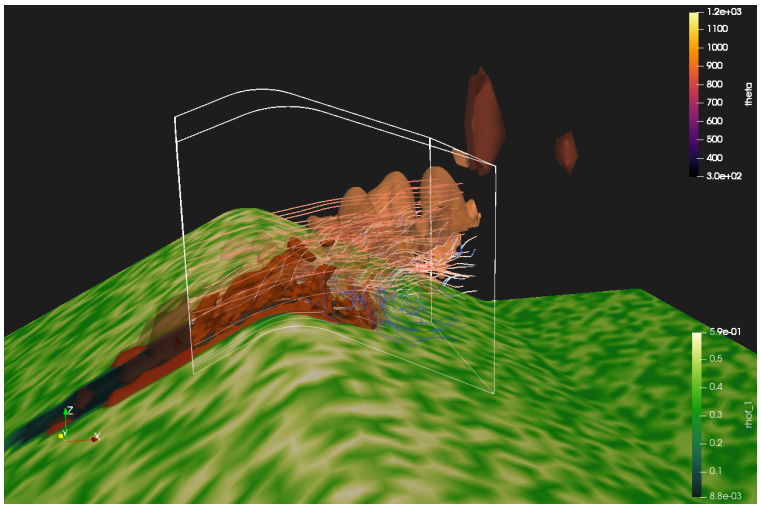
The same pattern applies whenever the time of an event in a remote series is unknown: a storm forming, two structures merging, a flow becoming unstable. An overview that keeps every timestep, cheap enough to take whole, lets the scientist find the event before committing to a sampling; a second request then fetches only the moment and the place that matter, at full resolution.

## References


[1] J.-L. Dupuy, R. R. Linn, V. Konovalov, F. Pimont, J. A. Vega, and E. Jiménez. Exploring three-dimensional coupled fire–atmosphere interactions downwind of wind-driven surface fires and their influence on backfires using the HIGRAD-FIRETEC model. *International Journal of Wildland Fire*, 20(6):734–750, 2011. doi: 10.1071/WF10035 1, 2

[2] R. Linn, J. Reisner, J. J. Colman, and J. Winterkamp. Studying wildfire behavior using FIRETEC. *International Journal of Wildland Fire*, 11(4):233–246, 2002. doi: 10.1071/WF02007 1

[3] B. Shneiderman. The eyes have it: a task by data type taxonomy for information visualizations. In *Proceedings 1996 IEEE Symposium on Visual Languages*, pp. 336–343, 1996. doi: 10.1109/VL.1996.545307 1




```
src  = source("ssh://host/wildfire/")
view = timesteps(src, 9000, 45000)
view = fields(view, ["theta", "u", "v", "w"])
view = region(view, x=(-100, 200), y=(-120, 120),
              z=(0, 350), coords="world")
save(view, "box.vts")
```

```
# from the provenance record in each output
Steps
  1. timesteps  keep timesteps #9000 to #45000
  2. fields     keep theta, u, v, w; drop O2, ...
  3. region     keep the samples inside the box
                (world coordinates), placed on
                the file's geometry
  4. save       write one VTK StructuredGrid file
                per timestep: each point at its
                own position from the source
```

Figure 3: A fire crossing a ridge, from a 96.6 GB simulation on a remote server, opened in ParaView. Outside the white box: Sieve's overview, every timestep at every eighth grid point (47 MB). Inside it: the time window and the region the scientist chose on that overview, fetched at full resolution (0.36 GB) by the request shown (top right). Sieve explains each output in a provenance record (bottom right, excerpt). The overview and the box are both saved with the position of every grid point, so ParaView places the box inside the overview with no alignment step. **[TODO: confirm the rendered timestep and the encodings: terrain colored by fuel left, the fire as a temperature surface, lines inside the box as streamlines of the full-resolution wind.]**